

Quantifying Fire Island, NY Coastal Erosion/Accretion from 2005-2024 With Remote Sensing and Image Classification Techniques

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Abstract:

This study evaluates long-term coastal change on Fire Island, New York, from 2005 to 2024 using satellite-based remote sensing techniques. Landsat 5, Landsat 8, and Sentinel-2 imagery were processed and analyzed through Modified Normalized Difference Water Index (MNDWI) classification to accurately differentiate water and land across three time points. All data were resampled to a consistent 30-meter resolution and clipped to a custom digitized boundary reflecting the 2005 historical coastline. Binary land and water classifications were generated by thresholding MNDWI outputs, and raster subtraction was performed to quantify shoreline change across two intervals (2005–2015 and 2015–2024) as well as the full study period.

The results reveal substantial erosion between 2005 and 2015, with a net loss of 10.7 km² of land, followed by significant recovery between 2015 and 2024, where 10.0 km² of land were regained. Over the entire nineteen-year period, Fire Island exhibited near-stability with a small net land gain of 0.2 km². While short-term fluctuations were considerable, the island demonstrated a high degree of resilience over decadal timescales, likely influenced by both natural processes and human interventions such as beach nourishment. This research highlights the effectiveness of open-access satellite data and simple spectral indices for coastal change detection. The methodology provides a scalable framework for monitoring other vulnerable barrier island systems facing sea level rise and intensifying storm impacts.

Introduction:

Coastal erosion is an increasingly urgent concern for many coastal communities, driven by a combination of natural processes and anthropogenic impacts. This project focuses on quantifying shoreline change along Fire Island, located off the south shore of Long Island, New York, over a nineteen-year period (2005–2024) using remote sensing methods like image classification, MNDWI, and change detection in ArcGIS Pro. Specifically, the research seeks to answer the following question: To what extent has coastal erosion and/or accretion changed the coast of Fire Island between 2005 and 2024?

Methodology:

- Acquired data through Landsat 5 and Landsat 8, as well as Sentinel-2
- Clipped imagery to custom Fire Island shapefile based on 2005 NIR band
- Resampled all datasets to 30-meter resolution to ensure consistency.
- Calculated the Modified Normalized Difference Water Index (MNDWI) for each year:

$$MNDWI = \frac{Green - SWIR}{Green + SWIR}$$

- Applied a conditional statement to prevent divide-by-zero errors.
- Thresholded MNDWI rasters at 0 to classify pixels as water (1) or land (0)
- Subtracted classified rasters between time periods (2005–2015, 2015–2024, and 2005–2024).
- Identified areas of erosion (land lost) and accretion (land gained).
- Multiplied pixel counts by 900 m² to calculate total land lost and gained.
- Converted results to square kilometers for reporting.

The change detection analysis revealed that Fire Island experienced notable shifts in coastal land area over the nineteen-year study period. Between 2005 and 2015, approximately 11.3 km² of land were lost to erosion while only 0.6 km² were gained through accretion, resulting in a net loss of 10.7 km². However, between 2015 and 2024, this trend reversed, with the island gaining 12.9 km² of land and losing only 2.9 km², yielding a net gain of 10.0 km². When viewed across the entire 2005–2024 timeframe, Fire Island showed near-equilibrium, with a slight net land gain of 0.2 km². These results highlight Fire Island's dynamic yet resilient nature, with significant short-term variability balanced by longer-term coastal stability.

Results:

The results of this study highlight the dynamic behavior of Fire Island's coastline, demonstrating substantial short-term variability alongside long-term resilience. Although the island experienced major erosion between 2005 and 2015, the following decade saw significant recovery, likely influenced by natural sediment transport and human interventions such as beach nourishment projects. The near-equilibrium land area observed between 2005 and 2024 suggests that, while Fire Island is vulnerable to episodic storm impacts, it retains a strong capacity for natural adjustment when restoration efforts support sediment availability. Minor classification errors due to cloud shadows and vessel wakes were present but did not materially affect the overall conclusions. The use of MNDWI proved effective for improving shoreline detection in sediment-rich coastal environments, offering a reliable and scalable approach for future monitoring of barrier island systems under rising sea levels and increasing storm intensities.

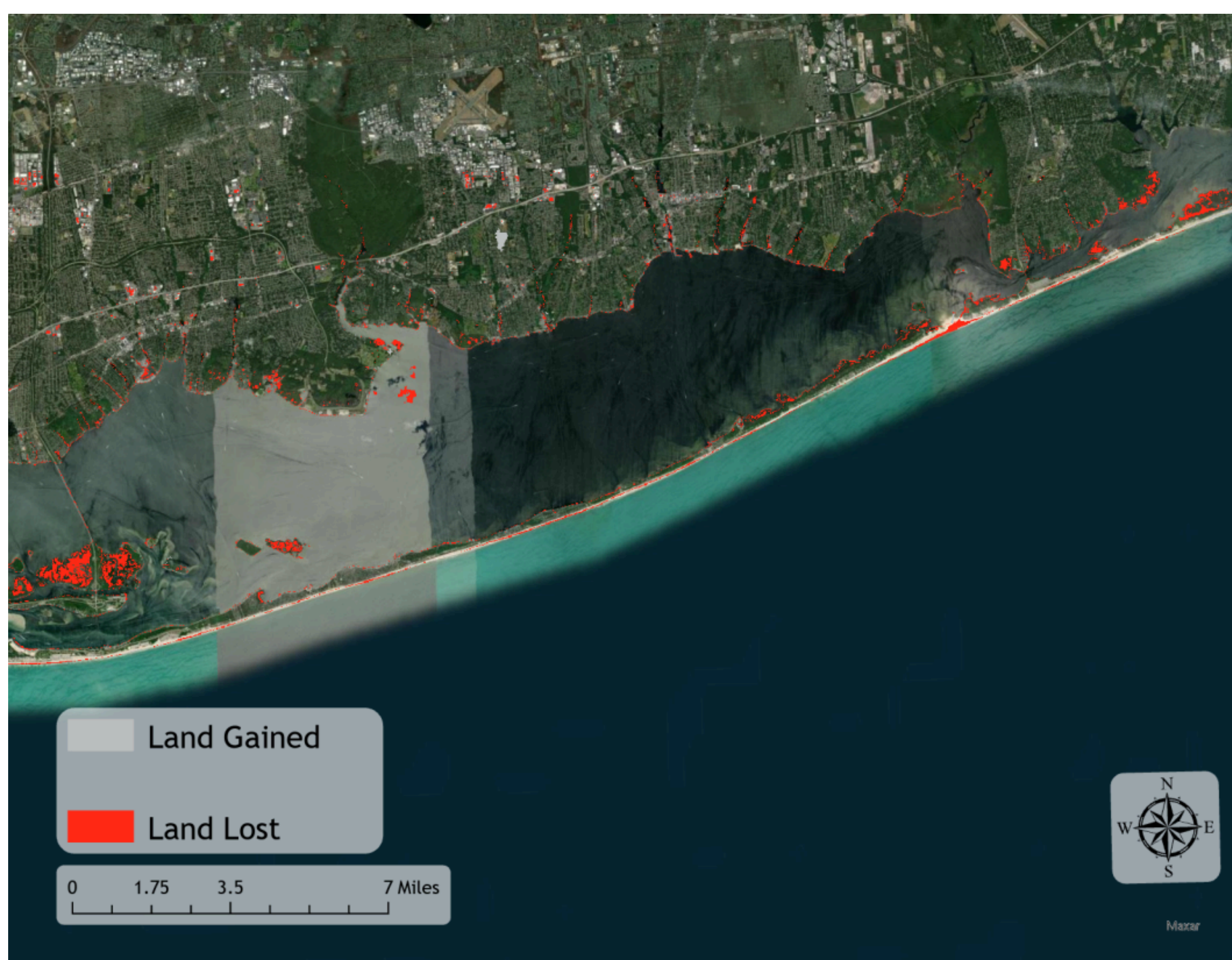


Figure 1: Quantified coastal change between 2005 and 2015

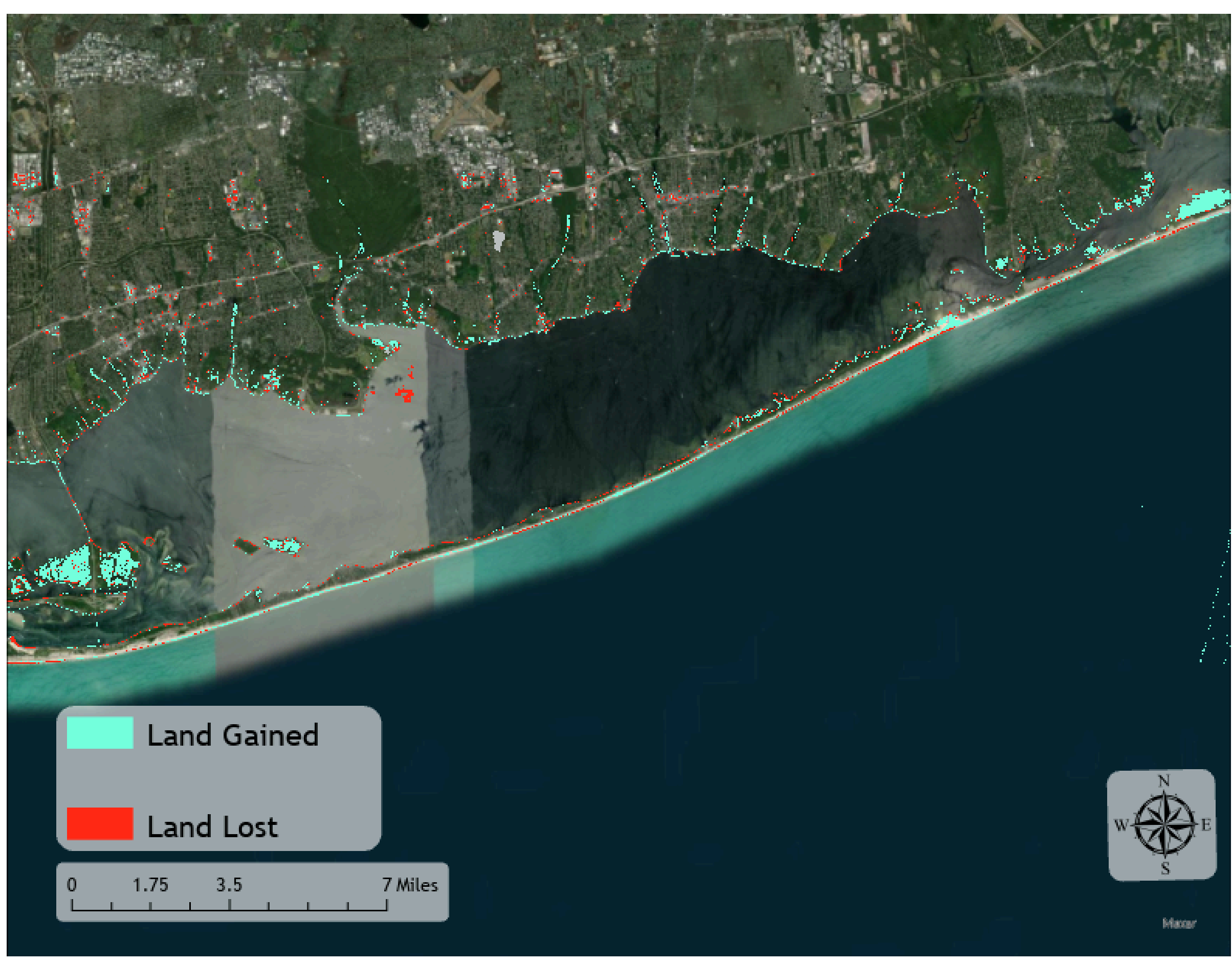


Figure 2: Quantified coastal change between 2015 and 2024

	2005-2015	2015-2024	2005-2024
Total Land Loss (square kilometers)	11.3	2.9	5.4
Total Land Gain (square kilometers)	0.6	12.9	5.6
Overall Land Change (square kilometers)	-10.7	+10.0	+0.2

Figure 3: Table of values showing the amount of land gained/lost in a given time interval. Measured in square kilometers.